Chapter 3. Coordinate Rings

Shangren Lu

March 22, 2026

Abstract

These notes develop affine algebraic geometry through the sequence

$$\boxed{X \longmapsto K[X] \longmapsto \mathrm{Spec}(K[X]).}$$

Starting from a classical affine algebraic set $X \subseteq \mathbb{A}^n_K$, we pass to its coordinate ring $K[X]$ and then to the prime spectrum. This progression distinguishes three settings: reducible classical affine geometry ($\mathrm{AffAlgSet}_K$), the irreducible subcategory (AffVar_K), and the scheme-theoretic extension (AffSch). Nullstellensatz identifies classical closed points with maximal ideals, whereas prime ideals parametrize irreducible closed subsets through generic points; thus the passage from maximal spectrum to prime spectrum is structurally necessary. Localization on $D(f)$ and pullback of functions provide the local and functorial mechanisms. The discussion culminates in the anti-equivalences

$$\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg,red}}, \qquad \mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}.$$

The organizing principle is functorial: polynomial rings are free objects, quotients impose relations by universal properties, localizations invert elements universally, and affine schemes are characterized by representability statements for ring-valued morphisms.

Contents

1 Roadmap **3**

2 Standing conventions and terminology **3**
2.1 Classical affine setting 3
2.2 Why category language helps here 4

3 Functions first: from equations to intrinsic algebra **4**
3.1 Why functions are fundamental 4
3.2 Polynomial algebras as free coordinate objects 4
3.3 Polynomial functions and coordinate rings 4
3.4 Running example: the parabola 6

4 Classical points, maximal ideals, and Nullstellensatz **6**
4.1 Points as evaluation maps 6

5 Why spectra are necessary: prime ideals as geometric points 7
5.1 From maximal ideals to all prime ideals 7
5.2 Zariski topology 8

6 Localization and basic opens 9
6.1 Preliminaries on sheaves and affine schemes 9
6.2 Algebraic setup and notation 10
6.3 Universal inversion and topology 10
6.4 Structure sheaf on basic opens 11

7 From algebra back to geometry 11
7.1 Classical object-level correspondence 11
7.2 Reverse-direction example: recover equations from a ring 11

8 Morphisms and contravariance 12
8.1 Contravariance and pullback of functions 12
8.2 Universal property of Spec and affine representability 13
8.3 Finite limits in affine schemes 13

1 Roadmap

1. Begin with a classical affine algebraic set and adopt a function-theoretic viewpoint.
2. Justify coordinate functions via the free K-algebra universal property of
$$K[x_1, \dots, x_n].$$
3. Construct the coordinate ring by universally imposing the relations vanishing on X:
$$K[X] \cong K[x_1, \dots, x_n]/I(X).$$
4. Recover classical points via evaluation and Nullstellensatz:
$$X(K) \cong \mathrm{MaxSpec}(K[X]).$$
5. Pass from maximal ideals to prime ideals in order to represent irreducible geometry by generic points.
6. Interpret localization through its universal property: R_f is the universal R-algebra in which f becomes invertible, corresponding to $D(f)$.
7. Conclude with contravariance, Hom-bijections, and the two affine dictionaries
$$\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg,red}}, \qquad \mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}.$$

2 Standing conventions and terminology

2.1 Classical affine setting

Throughout the classical part:

- K is an algebraically closed field;
- $X \subseteq \mathbb{A}^n_K$ is a Zariski-closed subset.

Definition 2.1 (Affine algebraic set)**.** A subset $X \subseteq \mathbb{A}^n_K$ cut out by polynomial equations is called an *affine algebraic set*.

Remark 2.2 (Terminology fix: "variety")**.** In these notes, the default classical category is $\mathrm{AffAlgSet}_K$, allowing reducible affine algebraic sets. Its algebraic counterpart is $\mathrm{Alg}_{K,\mathrm{fg,red}}$, the finitely generated reduced K-algebras. The label "affine variety" is reserved for the irreducible case; one then passes to AffVar_K, whose algebraic counterpart is $\mathrm{Alg}_{K,\mathrm{fg,dom}}$.

Remark 2.3 (Three affine levels kept in view)**.** We move between three compatible settings:

- classical reducible affine geometry: $\mathrm{AffAlgSet}_K \leftrightarrow \mathrm{Alg}_{K,\mathrm{fg,red}}$;
- classical irreducible affine geometry: $\mathrm{AffVar}_K \leftrightarrow \mathrm{Alg}_{K,\mathrm{fg,dom}}$;
- affine schemes over arbitrary rings: $\mathrm{AffSch}^{\mathrm{op}} \simeq \mathrm{CRing}$.

Only the first two require an algebraically closed base field; the third is the structural framework behind both.

2.2 Why category language helps here

Category-theoretic language is used to formulate uniqueness and functoriality precisely. The basic constructions are characterized, up to unique isomorphism, by universal mapping properties: $K[x_1, \ldots, x_n]$ represents $A \mapsto A^n$ on commutative K-algebras, R/I is initial among R-algebras on which I acts trivially, $S^{-1}R$ (in particular R_f) is initial among R-algebras in which S becomes invertible, and $\mathrm{Spec}(R)$ is characterized by representability of the corresponding mapping functor. This viewpoint provides a uniform passage between algebraic constructions and geometric interpretations.

3 Functions first: from equations to intrinsic algebra

3.1 Why functions are fundamental

Neither an embedding $X \hookrightarrow \mathbb{A}^n_K$ nor a generating set for its defining ideal is canonical. The ring of polynomial functions on X is intrinsic: it records polynomial observables on X together with their algebraic operations. Accordingly, coordinate rings are not auxiliary data but an algebraic presentation of the geometry itself, obtained from polynomial coordinate algebras by imposing the vanishing relations.

3.2 Polynomial algebras as free coordinate objects

Proposition 3.1 (Free commutative K-algebra on n generators)**.** *For every commutative K-algebra A and every n-tuple $(a_1, \ldots, a_n) \in A^n$, there is a unique K-algebra map*

$$\mathrm{ev}_{(a_1,\ldots,a_n)} : K[x_1, \ldots, x_n] \longrightarrow A$$

with $x_i \mapsto a_i$ for all i. Equivalently,

$$\boxed{\mathrm{Hom}_{K\text{-}alg}(K[x_1, \ldots, x_n], A) \cong A^n.}$$

The commutative triangle below records this universal property; the dashed arrow is uniquely determined by $x_i \mapsto a_i$.

$$\begin{tikzcd} K \arrow[r, "\iota"] \arrow[dr, "\eta_A"'] & K[x_1, \ldots, x_n] \arrow[d, dashed, "\exists!\ \mathrm{ev}_{(a_1,\ldots,a_n)}"] \\ & A \end{tikzcd}$$

Equivalently, $K[x_1, \ldots, x_n]$ represents the functor $A \mapsto A^n$ on commutative K-algebras.

Remark 3.2 (Naturality of polynomial coordinates)**.** The symbols $x_1, \ldots, x_n$ serve as universal coordinate functions: every tuple in a commutative K-algebra is obtained from them by a unique K-algebra specialization. This is the precise sense in which polynomial algebras provide canonical coordinates in the affine setting.

3.3 Polynomial functions and coordinate rings

Definition 3.3 (Polynomial function on X)**.** Let $X \subseteq \mathbb{A}^n_K$. A *polynomial function* on X is a function

$$X(K) \longrightarrow K, \qquad x \longmapsto f(x)$$

obtained by restricting some $f \in K[x_1, \ldots, x_n]$.

Definition 3.4 (Coordinate ring)**.** The *coordinate ring* of X is

$$K[X] := \{f|_X \,:\, f \in K[x_1, \ldots, x_n]\},$$

with pointwise ring operations.

Proposition 3.5 (Quotient rings impose relations universally)**.** *Let R be a commutative ring and $I \subseteq R$ an ideal. The quotient map $q : R \to R/I$ is characterized by the following universal property: for every commutative ring A and every homomorphism $\varphi : R \to A$ with $I \subseteq \ker \varphi$, there exists a unique map $\overline{\varphi} : R/I \to A$ such that*

$$\varphi = \overline{\varphi} \circ q.$$

Equivalently, R/I is initial among R-algebras on which every element of I acts by zero.

Under the condition $I \subseteq \ker \varphi$, the required factorization is represented by the commuting triangle below.

$$\begin{tikzcd} R \arrow[r, "q"] \arrow[dr, "\varphi"'] & R/I \arrow[d, dashed, "\exists!\,\overline{\varphi}"] \\ & A \end{tikzcd}$$

Equivalently, every ring map $R \to A$ annihilating I factors uniquely through R/I.

Proposition 3.6 (Coordinate ring as universal quotient)**.** *For $X \subseteq \mathbb{A}^n_K$, define*

$$I(X) := \{f \in K[x_1, \ldots, x_n] : f|_X = 0\}.$$

Then there is a natural isomorphism

$$\boxed{K[X] \cong K[x_1, \ldots, x_n]/I(X).}$$

Equivalently, $K[X]$ is obtained from the free coordinate algebra $K[x_1, \ldots, x_n]$ by imposing, in a universal manner, precisely the polynomial relations vanishing on X.

With quotient map q_X and kernel condition $\ker \rho = I(X)$, the first isomorphism theorem is encoded by the following commutative triangle.

$$\begin{tikzcd} K[x_1, \ldots, x_n] \arrow[r, "\rho"] \arrow[d, "q_X"'] & K[X] \\ K[x_1, \ldots, x_n]/I(X) \arrow[ur, "\sim"'] & \end{tikzcd}$$

Consequently, quotienting by $I(X)$ identifies two polynomials precisely when they induce the same function on X.

Proof. Let

$$\rho : K[x_1, \ldots, x_n] \to K[X], \qquad f \mapsto f|_X.$$

The map is surjective by definition of $K[X]$, and

$$\ker \rho = \{f : f|_X = 0\} = I(X).$$

Hence, by the first isomorphism theorem,

$$K[x_1, \ldots, x_n]/I(X) \cong K[X].$$

The universal characterization is the previous proposition applied to $R = K[x_1, \ldots, x_n]$ and $I = I(X)$. $\square$

Corollary 3.7 (Finite type and reducedness in the classical setting)**.** *If $X \subseteq \mathbb{A}^n_K$ is an affine algebraic set, then $K[X]$ is a finitely generated reduced K-algebra.*

Proof. Finite generation follows immediately from the quotient presentation. For reducedness, let $\bar{f} \in K[X]$ satisfy $\bar{f}^m = 0$ for some $m \geq 1$. Evaluating at any $x \in X(K)$ gives $\bar{f}(x)^m = 0$ in the field K, hence $\bar{f}(x) = 0$. Therefore $\bar{f} = 0$ as a function on X, so $K[X]$ is reduced. Equivalently, $I(X)$ is radical; over algebraically closed K, this is the identity $I(V(I(X))) = \mathrm{rad}(I(X))$ from Nullstellensatz. □

3.4 Running example: the parabola

Example 3.8 (Parabola)**.** Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2_K.$$

Then

$$K[X] \cong K[x, y]/(y - x^2) \cong K[x],$$

via substitution $y = x^2$. Hence the parabola and the affine line have isomorphic coordinate rings.

$$\boxed{K[V(y - x^2)] \cong K[x].}$$

Remark 3.9 (Classical notation $\Gamma(X)$)**.** In this classical affine setting one may write

$$\Gamma(X) = K[X],$$

meaning global regular functions are exactly polynomial functions on X.

4 Classical points, maximal ideals, and Nullstellensatz

4.1 Points as evaluation maps

Proposition 4.1 (Evaluation map)**.** *For $x \in X(K)$, evaluation defines a K-algebra map*

$$\mathrm{ev}_x : K[X] \to K, \qquad f \mapsto f(x).$$

Its kernel

$$\mathfrak{m}_x := \ker(\mathrm{ev}_x)$$

is a maximal ideal of $K[X]$.

The relation between evaluation and residue fields is summarized by the commuting triangle below, where $\mathfrak{m}_x = \ker(\mathrm{ev}_x)$.

$$\begin{tikzcd} K[X] \arrow[r, "\mathrm{ev}_x"] \arrow[d, "q_x"'] & K \\ K[X]/\mathfrak{m}_x \arrow[ur, "\sim"'] & \end{tikzcd}$$

Thus evaluation at a K-point agrees with passage to the residue field at the corresponding closed point.

Proof. The map is a unital K-algebra homomorphism and is surjective because constants lie in $K[X]$. Therefore

$$K[X]/\mathfrak{m}_x \cong K,$$

so $\mathfrak{m}_x$ is maximal. □

Theorem 4.2 (Hilbert's Nullstellensatz (forms used here))**.** *Let K be algebraically closed.*

(i) For $I \subseteq K[x_1, \dots, x_n]$, one has $I(V(I)) = \mathrm{rad}(I)$.

(ii) Maximal ideals of $K[x_1, \dots, x_n]$ are exactly

$$(x_1 - a_1, \dots, x_n - a_n), \qquad (a_1, \dots, a_n) \in K^n.$$

Applying Nullstellensatz through the quotient presentation of $K[X]$ yields the classical correspondence between points and maximal ideals. The map $x \mapsto \mathfrak{m}_x$ is injective because distinct points give distinct evaluation maps. For surjectivity, let $\mathfrak{m} \in \mathrm{MaxSpec}(K[X])$ and let $q : K[x_1, \dots, x_n] \twoheadrightarrow K[X]$ be the quotient map. Then $q^{-1}(\mathfrak{m})$ is maximal, so

$$q^{-1}(\mathfrak{m}) = (x_1 - a_1, \dots, x_n - a_n)$$

for some $(a_1, \dots, a_n) \in K^n$. The inclusion $I(X) \subseteq q^{-1}(\mathfrak{m})$ implies $(a_1, \dots, a_n) \in V(I(X))$. Since $X = V(I(X))$, one has $(a_1, \dots, a_n) \in X(K)$, and hence $\mathfrak{m} = \mathfrak{m}_{(a_1,\dots,a_n)}$.

$$\boxed{X(K) \cong \mathrm{Hom}_{K\text{-alg}}(K[X], K) \cong \mathrm{MaxSpec}(K[X]).}$$

Consequently, one obtains the chain

$$\boxed{X(K) \;\cong\; \mathrm{MaxSpec}(K[X]) \;\subseteq\; \mathrm{Spec}(K[X]).}$$

Remark 4.3 (Jacobson viewpoint)**.** Finite-type K-algebras are Jacobson rings, so closed points (maximal ideals) remain dense and highly informative even though Spec has extra non-closed points.

Example 4.4 (Affine line: classical points inside the spectrum)**.** Let $R = K[x]$. Its maximal ideals are precisely $(x - a)$ for $a \in K$, so

$$\mathrm{MaxSpec}(R) = \{(x - a) : a \in K\} \cong \mathbb{A}^1_K(K).$$

In $\mathrm{Spec}(R)$ there is one additional prime ideal, namely (0), and

$$\overline{\{(0)\}} = \mathrm{Spec}(K[x]).$$

Thus $\mathrm{Spec}(K[x])$ consists of the classical closed points together with the non-closed generic point (0).

5 Why spectra are necessary: prime ideals as geometric points

5.1 From maximal ideals to all prime ideals

The maximal spectrum recovers classical closed points, but irreducible closed subsets are not represented as points of MaxSpec. Passing to Spec adds precisely the generic points required to encode irreducible geometry.

Definition 5.1 (Prime spectrum)**.** For a commutative ring R,

$$\mathrm{Spec}(R) := \{\mathfrak{p} \subseteq R : \ \mathfrak{p} \text{ prime}\}.$$

5.2 Zariski topology

Definition 5.2 (Closed sets and standard opens)**.** For an ideal $I \subseteq R$, define

$$V(I) := \{\mathfrak{p} \in \operatorname{Spec}(R) : I \subseteq \mathfrak{p}\}.$$

These are the closed sets. For $f \in R$, define

$$D(f) := \operatorname{Spec}(R) \setminus V((f)) = \{\mathfrak{p} \in \operatorname{Spec}(R) : f \notin \mathfrak{p}\}.$$

The sets $D(f)$ form a basis of opens.

Proposition 5.3 (Closed points and closure of one point)**.** *For $\mathfrak{p} \in \operatorname{Spec}(R)$,*

$$\boxed{\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).}$$

Consequently, $\mathfrak{p}$ is a closed point if and only if $\mathfrak{p}$ is maximal.

Proof. A closed set $V(I)$ contains $\mathfrak{p}$ exactly when $I \subseteq \mathfrak{p}$. The smallest such closed set is $V(\mathfrak{p})$. It is a singleton exactly when no strictly larger prime contains $\mathfrak{p}$, equivalently when $\mathfrak{p}$ is maximal. □

Proposition 5.4 (Irreducible closed subsets and generic points)**.** *The assignment*

$$\mathfrak{p} \longmapsto V(\mathfrak{p})$$

induces a bijection

$$\boxed{\{\textit{prime ideals of } R\} \longleftrightarrow \{\textit{irreducible closed subsets of } \operatorname{Spec}(R)\}.}$$

Hence irreducible closed subsets are represented by prime ideals. Each irreducible closed subset has a unique generic point, namely its defining prime ideal.

Remark 5.5 (Specialization and generization)**.** For $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(R)$, one says that $\mathfrak{p}$ is a specialization of $\mathfrak{q}$ if and only if

$$\mathfrak{q} \subseteq \mathfrak{p},$$

and equivalently $\mathfrak{q}$ is a generization of $\mathfrak{p}$; this relation is precisely reverse inclusion in the poset of prime ideals.

Example 5.6 (Generic point of the parabola)**.** Let $R = K[x, y]$ and let

$$\mathfrak{p} := (y - x^2) \subseteq R.$$

Since

$$R/\mathfrak{p} \cong K[x]$$

is a domain, $\mathfrak{p}$ is prime. Therefore

$$Y := V(\mathfrak{p}) \subseteq \operatorname{Spec}(R)$$

is irreducible, and its generic point is exactly $\mathfrak{p}$. Equivalently,

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}) = Y.$$

Every closed point $\mathfrak{m}_{a,b} = (x - a, y - b)$ on Y satisfies $b = a^2$, so

$$\mathfrak{p} \subseteq (x - a, y - a^2),$$

showing each closed point of the parabola is a specialization of the generic point.

Example 5.7 (The affine plane in spectral language)**.** In $\mathrm{Spec}(K[x,y])$:

- Closed points of the affine plane are the maximal ideals $(x-a, y-b)$, i.e. classical points $(a,b) \in K^2$.
- Each irreducible plane curve contributes a generic point: for instance $(y-3x^2)$ is the generic point of $V(y-3x^2)$, and
$$\overline{\{(y-3x^2)\}} = V(y-3x^2).$$
- The ideal (0) is the generic point of the whole irreducible surface, with
$$\overline{\{(0)\}} = \mathrm{Spec}(K[x,y]).$$

Accordingly, spectra simultaneously encode closed points, irreducible curves via their generic points, and the generic point of the ambient irreducible space:

$$\boxed{\text{classical points} \;\leftrightarrow\; \text{maximal ideals} \;\subsetneq\; \text{all prime ideals.}}$$

6 Localization and basic opens

6.1 Preliminaries on sheaves and affine schemes

The scheme-theoretic viewpoint replaces a space by a pair consisting of an underlying topological space and rings of functions on its open subsets.

Definition 6.1 (Sheaf of rings)**.** Let X be a topological space. A *sheaf of rings* $\mathcal{F}$ on X consists of:

- a ring $\mathcal{F}(U)$ for each open set $U \subseteq X$;
- a restriction map $\rho^U_V : \mathcal{F}(U) \to \mathcal{F}(V)$ for each inclusion $V \subseteq U$,

satisfying identity and composition for restrictions, together with:

(i) *locality*: if $s, t \in \mathcal{F}(U)$ have equal restrictions on an open cover of U, then $s = t$;

(ii) *gluing*: compatible sections on an open cover of U glue uniquely to a section of $\mathcal{F}(U)$.

Definition 6.2 (Stalk and locally ringed space)**.** For $x \in X$, the *stalk* $\mathcal{F}_x$ is the direct limit of the rings $\mathcal{F}(U)$ over neighborhoods $U \ni x$, equivalently the ring of germs at x. A *locally ringed space* is a pair $(X, \mathcal{O}_X)$ such that each stalk $\mathcal{O}_{X,x}$ is a local ring.

Definition 6.3 (Affine scheme)**.** Let R be a commutative ring and $X := \mathrm{Spec}(R)$. Equipping X with the structure sheaf characterized on basic opens by
$$\mathcal{O}_X(D(f)) = R_f \qquad (f \in R),$$
one obtains the affine scheme $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$.

Example 6.4 (Punctured affine line)**.** Let $R = K[x]$ and $X := \operatorname{Spec}(K[x])$. For the basic open

$$D(x) = \{\mathfrak{p} \in \operatorname{Spec}(K[x]) : x \notin \mathfrak{p}\},$$

one has

$$\Gamma(D(x), \mathcal{O}_X) = K[x]_x = K[x, x^{-1}].$$

Hence x^{-1} is regular on $D(x)$ but not global on X, since

$$\Gamma(X, \mathcal{O}_X) = K[x].$$

This exhibits the local nature of regularity in scheme theory.

6.2 Algebraic setup and notation

Let R be a commutative ring and set $X := \operatorname{Spec}(R)$. For a multiplicative subset $S \subseteq R$, write

$$i_S : R \to S^{-1}R, \qquad r \longmapsto \frac{r}{1},$$

for the localization map. For $f \in R$, put

$$S_f := \{1, f, f^2, \dots\}, \qquad R_f := S_f^{-1}R, \qquad i_f := i_{S_f}.$$

For any ring A, the notation $A^\times$ denotes its group of units. Recall also

$$D(f) := \{\mathfrak{p} \in \operatorname{Spec}(R) : f \notin \mathfrak{p}\}.$$

Proposition 6.5 (Basic opens form a basis)**.** *For $f, g \in R$,*

$$D(1) = \operatorname{Spec}(R), \qquad D(0) = \varnothing, \qquad D(fg) = D(f) \cap D(g).$$

Consequently, the sets $D(f)$ form a basis of the Zariski topology on $\operatorname{Spec}(R)$.

Proof. The first two identities are immediate from the definitions. For the third identity,

$$\mathfrak{p} \in D(fg) \iff fg \notin \mathfrak{p} \iff f \notin \mathfrak{p} \text{ and } g \notin \mathfrak{p} \iff \mathfrak{p} \in D(f) \cap D(g),$$

using primality of $\mathfrak{p}$. □

6.3 Universal inversion and topology

Proposition 6.6 (Localization as universal inversion)**.** *Let R be a commutative ring and $S \subseteq R$ a multiplicative subset. The canonical map $i_S : R \to S^{-1}R$ is characterized by:*

- *each $i_S(s)$ ($s \in S$) is invertible in $S^{-1}R$;*
- *for any ring map $\varphi : R \to A$ such that $\varphi(S) \subseteq A^\times$, there exists a unique map $\widetilde{\varphi} : S^{-1}R \to A$ with $\varphi = \widetilde{\varphi} \circ i_S$.*

In particular, R_f is the universal R-algebra in which f becomes invertible.

For $\varphi : R \to A$ with $\varphi(f) \in A^\times$, the universal factorization is encoded by

$$\begin{tikzcd} R \arrow[r, "i_f"] \arrow[dr, "\varphi"'] & R_f \arrow[d, dashed, "\exists!\,\widetilde{\varphi}"] \\ & A \end{tikzcd}$$

Proposition 6.7 (Basic opens as localized spectra)**.** *For $f \in R$, the map $i_f : R \to R_f$ induces a homeomorphism*

$$\mathrm{Spec}(R_f) \xrightarrow{\sim} D(f), \qquad \mathfrak{q} \longmapsto i_f^{-1}(\mathfrak{q}).$$

Its inverse sends $\mathfrak{p} \in D(f)$ to the extended ideal $\mathfrak{p}R_f$.

Proof. Prime ideals of R_f correspond to prime ideals of R disjoint from S_f. Since $S_f = \{1, f, f^2, \dots\}$, this is equivalent to $f \notin \mathfrak{p}$, i.e.

$$\mathfrak{p} \in D(f).$$

Under this correspondence, contraction and extension are mutually inverse and preserve inclusions, hence induce the stated homeomorphism. □

6.4 Structure sheaf on basic opens

Definition 6.8 (Structure sheaf on an affine spectrum)**.** The structure sheaf $\mathcal{O}_X$ is the sheaf of rings determined on the basis of basic opens by

$$\mathcal{O}_X(D(f)) := R_f, \qquad f \in R,$$

with restriction morphisms induced by localization. For any open subset $U \subseteq X$, write $\Gamma(U, \mathcal{O}_X)$ for the ring of sections on U.

Proposition 6.9 (Sections on basic opens)**.** *For $f \in R$,*

$$\boxed{\Gamma(D(f), \mathcal{O}_X) \cong R_f.}$$

Equivalently, regular functions on $D(f)$ are exactly fractions a/f^n with $a \in R$ and $n \geq 0$.

7 From algebra back to geometry

7.1 Classical object-level correspondence

Theorem 7.1 (Classical affine dictionary)**.** *Over algebraically closed K, there is an anti-equivalence of categories*

$$\boxed{\mathrm{AffAlgSet}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg,red}}.}$$

If one restricts to irreducible objects, this refines to

$$\mathrm{AffVar}_K^{\mathrm{op}} \simeq \mathrm{Alg}_{K,\mathrm{fg,dom}}.$$

7.2 Reverse-direction example: recover equations from a ring

Example 7.2. Let

$$A := K[u^2, uv, v^2] \subseteq K[u, v].$$

Define a surjective map

$$\psi : K[x, y, z] \twoheadrightarrow A, \qquad x \mapsto u^2,\ y \mapsto uv,\ z \mapsto v^2.$$

The relation

$$xz - y^2 = 0$$

holds in A, and in fact

$$\ker \psi = (xz - y^2).$$

Hence

$$A \cong K[x, y, z]/(xz - y^2).$$

Therefore the associated classical affine algebraic set is

$$V(xz - y^2) \subseteq \mathbb{A}^3_K,$$

and the associated affine scheme is

$$\operatorname{Spec}(A) \cong \operatorname{Spec}\big(K[x, y, z]/(xz - y^2)\big).$$

8 Morphisms and contravariance

8.1 Contravariance and pullback of functions

A morphism $f : X \to Y$ induces, by precomposition, a homomorphism of function rings

$$f^* : K[Y] \to K[X], \qquad g \mapsto g \circ f.$$

Therefore the coordinate-ring construction is contravariant in the geometric variable. For schemes this yields the global-sections functor

$$\Gamma : \mathrm{AffSch}^{\mathrm{op}} \longrightarrow \mathrm{CRing}, \qquad X \longmapsto \Gamma(X, \mathcal{O}_X).$$

On affine schemes, this contravariant functor is paired with Spec in the affine anti-equivalence.

Definition 8.1 (Pullback on coordinate rings)**.** For a morphism $f : X \to Y$ of affine algebraic sets, define

$$f^* : K[Y] \to K[X], \qquad g \mapsto g \circ f.$$

The arrow reversal is visible in the following commutative evaluation square for each $x \in X(K)$.

$$\begin{tikzcd} K[Y] \arrow[r, "f^*"] \arrow[d, "\mathrm{ev}_{f(x)}"'] & K[X] \arrow[d, "\mathrm{ev}_x"] \\ K \arrow[r, equal] & K \end{tikzcd}$$

Proposition 8.2 (A ring map induces a map of spectra)**.** *A homomorphism $\varphi : R \to S$ induces a continuous map*

$$\operatorname{Spec}(S) \to \operatorname{Spec}(R), \qquad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

8.2 Universal property of Spec and affine representability

Theorem 8.3 (Universal property of affine target)**.** *For every locally ringed space X and commutative ring R, there is a natural bijection*

$$\boxed{\mathrm{Hom}_{\mathrm{LRS}}(X, \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, \Gamma(X, \mathcal{O}_X)).}$$

Equivalently, $\mathrm{Spec}(R)$ *represents the functor*

$$X \longmapsto \mathrm{Hom}_{\mathrm{CRing}}(R, \Gamma(X, \mathcal{O}_X))$$

on locally ringed spaces.

For a ring map $\alpha : R \to \Gamma(X, \mathcal{O}_X)$, the corresponding morphism $u_\alpha : X \to \mathrm{Spec}(R)$ is characterized by the commuting triangle below.

$$\begin{tikzcd} R \arrow[r, "\alpha"] \arrow[d, "\cong"'] & \Gamma(X, \mathcal{O}_X) \\ \Gamma(\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)}) \arrow[ur, "\Gamma(u_\alpha)"'] & \end{tikzcd}$$

Accordingly, a morphism into an affine scheme is determined by the induced homomorphism on global sections.

Theorem 8.4 (Affine schemes Hom-bijection)**.** *For commutative rings R, S,*

$$\boxed{\mathrm{Hom}_{\mathrm{AffSch}}(\mathrm{Spec}(S), \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, S).}$$

Proof. Apply the previous theorem to $X = \mathrm{Spec}(S)$, then use

$$\Gamma(\mathrm{Spec}(S), \mathcal{O}_{\mathrm{Spec}(S)}) \cong S.$$

□

Theorem 8.5 (Classical affine Hom-bijection)**.** *For affine algebraic sets X, Y over algebraically closed K,*

$$\boxed{\mathrm{Hom}_{\mathrm{AffAlgSet}_K}(X, Y) \cong \mathrm{Hom}_{K\text{-}alg}(K[Y], K[X]).}$$

Remark 8.6 (Affine anti-equivalence)**.** The affine-scheme Hom-bijection yields the central structural statement: $\mathrm{Spec} : \mathrm{CRing}^{\mathrm{op}} \to \mathrm{AffSch}$ and $\Gamma : \mathrm{AffSch}^{\mathrm{op}} \to \mathrm{CRing}$ are quasi-inverse equivalences. The classical dictionary is the finite-type reduced K-algebra slice of this statement.

8.3 Finite limits in affine schemes

Proposition 8.7 (Products and fibre products)**.** *Affine schemes admit finite limits; in particular, products and fibre products are given by*

$$\boxed{\mathrm{Spec}(R) \times_{\mathrm{Spec}(\mathbb{Z})} \mathrm{Spec}(S) \cong \mathrm{Spec}(R \otimes_{\mathbb{Z}} S)}$$

and, for R-algebras A, B,

$$\boxed{\mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(B) \cong \mathrm{Spec}(A \otimes_R B).}$$

The fibre-product formula is encoded by the pullback square below.

$$\begin{tikzcd} \mathrm{Spec}(A \otimes_R B) \arrow[r, "\mathrm{pr}_B"] \arrow[d, "\mathrm{pr}_A"'] & \mathrm{Spec}(B) \arrow[d] \\ \mathrm{Spec}(A) \arrow[r] & \mathrm{Spec}(R) \end{tikzcd}$$

This identifies tensor product with affine base change on spectra.

Proof. Tensor products are characterized by the following mapping properties, natural in the target ring:

$$\mathrm{Hom}_{\mathrm{CRing}}(R \otimes_{\mathbb{Z}} S, T) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, T) \times \mathrm{Hom}_{\mathrm{CRing}}(S, T),$$

$$\mathrm{Hom}_{R\text{-alg}}(A \otimes_R B, T) \cong \mathrm{Hom}_{R\text{-alg}}(A, T) \times \mathrm{Hom}_{R\text{-alg}}(B, T).$$

Under the affine Hom-bijection, these are exactly the universal properties of products and fibre products in AffSch. □

Remark 8.8 (Functor-of-points viewpoint)**.** For an affine scheme $\mathrm{Spec}(R)$,

$$\boxed{h_{\mathrm{Spec}(R)}(A) := \mathrm{Hom}_{\mathrm{AffSch}}(\mathrm{Spec}(A), \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathrm{CRing}}(R, A).}$$

Hence $h_{\mathrm{Spec}(R)}$ is representable, and the ring R is recovered from its functor of points.